# TD n° 1.

Ex 1. Rocq

Ex 2. Rocq

Ex 3. If $\Gamma \vdash A$ is provable in NJ, take some proof tree in NJ for it, it is a valid proof tree in NJ + R, thus $\Gamma \vdash A$ is provable in NJ + R.

Conversely, take a proof tree $\pi$ of $\Gamma \vdash A$ in NJ + R, by admissibility of R, every occurence of R in $\pi$ can be replaced by part of a tree that doesn't use R. Doing this starting from the leaves yield a valid proof tree for $\Gamma \vdash A$ in NJ.

Ex 4.

1. Take a proof tree of $\Gamma, A, B \vdash C$. Replace each occurence of

$$\frac{}{\Gamma, A, B, \Delta \vdash A}\,Ax \qquad \frac{\dfrac{}{\Gamma, A\wedge B, \Delta \vdash A \wedge B}\,Ax}{\Gamma, A \wedge B, \Delta \vdash A}\,\wedge eg$$

and

$$\frac{}{\Gamma, A, B, \Delta \vdash B}\,Ax$$

with

$$\frac{\dfrac{}{\tilde{\Delta} \vdash A \wedge B}\,Ax}{\tilde{\Delta} \vdash B}\,\wedge ed,$$

also replacing every sequent $\Gamma, A, B, \Delta \vdash D$ with $\Gamma, A \wedge B, \Delta \vdash D$.

2. By induction on the size of $\Gamma$, we apply Q1.

3. No, derivable rules cannot remove hypotheses (with the rules in NJ).

# Ex 5.

1. $\dfrac{\Xi,\Gamma\vdash A}{\Xi,\Delta\vdash B}$ admissible iff $(\Xi,\Gamma\vdash A$ implies $\Xi,\Delta\vdash B)$ } Ex 4.

iff $(\Xi\vdash\bigwedge\Gamma\Rightarrow A$ implies $\Xi\vdash\bigwedge\Delta\Rightarrow B)$

iff $\Xi\vdash(\bigwedge\Gamma\Rightarrow A)\Rightarrow(\bigwedge\Delta\Rightarrow B)$ } currying

iff $\Xi\vdash(\bigwedge\Gamma\Rightarrow A\wedge\bigwedge\Delta\Rightarrow B$

2. Similarly to Q1, we know

$$\Xi,\Gamma\vdash(\bigwedge\Delta\Rightarrow A)\Rightarrow B$$

is provable, thus, by repeated currying, writing $\Delta=F_1,\dots,F_n$, the sequent

$$\Xi,\Gamma\vdash\underbrace{(F_1\Rightarrow\cdots\Rightarrow F_n\Rightarrow A)}_{\Phi}\Rightarrow B$$

is provable; take $\Pi$ a proof tree for it.

$$\dfrac{\dfrac{\Pi}{\Xi,\Gamma\vdash\Phi\Rightarrow B}\qquad\dfrac{\dfrac{}{\Xi,\Gamma,\Delta\vdash A}\text{assumption}}{\Xi,\Gamma\vdash F_1\Rightarrow\cdots\Rightarrow F_n\Rightarrow A}\text{repeated }\Rightarrow_I}{\Xi,\Gamma\vdash B}\Rightarrow_E$$

# Ex 6.

1.

$$\dfrac{\dfrac{\dfrac{\Pi_1}{\Gamma\vdash A}}{\Gamma,\Delta\vdash A}wk\qquad\dfrac{\dfrac{\Pi_2}{\Delta\vdash B}}{\Gamma,\Delta\vdash B}wk}{\Gamma,\Delta\vdash A\wedge B}\wedge I$$

2.

$$\dfrac{\dfrac{\Gamma\vdash\neg A}{\Gamma,\Delta\vdash\neg A}wk\qquad\dfrac{\Delta\vdash A}{\Gamma,\Delta\vdash A}wk}{\Gamma,\Delta\vdash\bot}\neg E$$

$$\dfrac{\dfrac{\Gamma\vdash A\vee B}{\Gamma,\Delta,\Sigma\vdash A\vee B}wk\qquad\dfrac{\Delta,A\vdash C}{\Gamma,\Delta,\Sigma,A\vdash C}wk\qquad\dfrac{\Sigma,B\vdash C}{\Gamma,\Delta,\Sigma,B\vdash C}wk}{\Gamma,\Delta,\Sigma\vdash C}\vee E$$

$$\dfrac{\dfrac{\Gamma \vdash A \Rightarrow B}{\Gamma, \Sigma \vdash A \Rightarrow B}\,wk \qquad \dfrac{\Delta \vdash B}{\Gamma, \Delta \vdash B}\,wR}{\Gamma, \Sigma \vdash B}\Rightarrow E.$$

Ex 7.

Define $C_\top := \bot \Rightarrow \bot$ and $C_{\neg A} := A \Rightarrow \bot$.

$$\dfrac{\dfrac{}{\Gamma, \bot \vdash \bot}\,Ax}{\Gamma \vdash C_\top}\Rightarrow I \quad \text{"}\top I\text{"}$$

$$\dfrac{\Gamma \vdash C_{\neg A} \qquad \Gamma \vdash A}{\Gamma \vdash \bot}\Rightarrow E$$

$\neg E.$

$$\dfrac{\Gamma, A \vdash \bot}{\Gamma \vdash C_{\neg A}}\Rightarrow I \quad \neg I$$

Ex 8. Rocq

Ex 9. Rocq

$\neg_R A$ is equivalent to $\neg A$ iff $\vdash \neg R$ is provable.

TD n° 2.

# I) Preliminaries

## Exercise 1.

Take $R$ a relation over $\lambda$-terms.
Then for all $M \to_R N$, for all $C[\,]$, we have $C[M] \to_R C(N)$

by def. of $\to_R$.

## Exercise 2

By induction on $M R^* N$,
if $M \; R^0 \; N$, i.e. $M = N$, then $C[M] \; R^0 \; C[M] = C[N]$.

if $M \; R \; N \; R^* \; P$ then by ih $C[N] \; R^* \; C[P]$

and $C[M] \; R \; C[N]$ by congruence.

# II) The complexity of the simply-typed $\beta$-reduction.

## Exercise 3.

For every $n \in \mathbb{N}$,

$$D_{A_n} = \lambda f_{n+1} : A_n \to A_n . \lambda f_n : A_n, \underbrace{f_{n+1} \underbrace{(f_{n+1} \; f_n)}_{: A_n}}_{: A_n} : \underbrace{A_{n+1} \to A_{n+1}}_{= A_{n+2}}$$

By induction on $k \in \mathbb{N}$, we show

$$T_{A_n, k} : A_{n+2} \to A_{n+3} \to \cdots \to A_{n+2+k}$$

$$T_{A_{n,0}} = D_{A_n} : A_{n+2}$$

$$T_{A_n, k+1} = T_{A_{n+1}, k} \; \underset{A_{n+2}}{\underbrace{D_{A_n}}}$$

Exercise 4.

By induction on $k$,

- $T_{A_n,0} = D_{A_n}$ is already in $\beta$-normal form

$$\lambda f_{n+1}.\lambda f_n.\, f_{n+1}^{2}\ f_n.$$

as $\sigma(1)=2$

- $T_{A_n,k+1} = T_{A_{n+1},k}\, D_{A_n} \longrightarrow_\beta^* (\lambda f_{n+2}.\lambda f_{n+1}.\, f_{n+2}^{\sigma(k+1)}\ f_{n+1})$

$$(\lambda f_{n+1}.\lambda f_n.\, f_{n+1}^{2}\ f_n)$$

$$\longrightarrow_\beta \lambda f_{n+1}.(\lambda f_{n+1}.\lambda f_n.\, f_{n+1}^{2}\ f_n)^{\sigma(k+1)}\ f_{n+1}$$

$$\longrightarrow_\beta^* \lambda f_{n+1}.\lambda f_n.\, f_{n+1}^{2^{\sigma(k+1)}}\ f_n.$$

## III | Taming $\beta$-reduction using the size of types.

Exercise 5. By induction on $M$.

$x \longrightarrow_\beta^* x = x^*$ $\quad (\lambda x. M) \longrightarrow_\beta^* \lambda x. M^* = (\lambda x. M)^*$ $\quad M\,N \longrightarrow_\beta^* M^*\,N \longrightarrow_\beta^* M^*\,N^*$

$(\lambda x. L)\,N \longrightarrow_\beta^* (\lambda x. L^*)\,N \longrightarrow_\beta^* (\lambda x. L^*)\,N^* \longrightarrow_\beta L^*\{N^*/x\}.$

Exercise 6.

$$((\lambda x.\, x\, y)\,(\lambda a.a))^* = (x\,y)\ \{\lambda a.a/x\}$$
$$= (\lambda a.a)\,y \longrightarrow_\beta y$$

so not a $\beta$-normal form.

None of the $\beta$-redexes in $M$ are still present in $M^*$.

## Exercise 7.

$$\omega_m(M\{N/x\}) = \max\{\omega(R) \mid R \text{ is a } \beta\text{-redex in } M\{N/x\}\}$$

$$= \max\Big(\{\omega(R) \mid R \text{ is a } \beta\text{-redex in } M\}$$

$$\cup\{\omega(R) \mid R \text{ is a } \beta\text{-redex in } N\}$$

$$\cup\Big\{\omega((\lambda x.U)\, a) \;\Big|\; \begin{array}{l}(\lambda x.U) = N \\ x\, a \in M\end{array}\Big\}\Big)$$

$$= \max(\omega_m(M), \omega_m(N), size(A)).$$

We can prove this by induction on $M$:

- $\omega_m(x\{N/x\}) = \omega_m(N) \leq \max(\dots)$
- $\omega_m((\lambda y.M)\{N/x\}) = \omega_m(\lambda y.(M\{N/x\})) = \omega_m(M\{N/x\}) \leq \max(\dots)$
- $\omega_m((x\; M)\{\lambda y.N/x\}) = \omega_m((\lambda y.N)(M\{N/x\}))$
  $\leq \max(size(A), \omega_m(M), \omega_m(N))$

  $(M, N) \neq (x\, U, \lambda y.V)$
- $\omega_m(MM'\{N/x\}) = \omega_m((M\{N/x\})(M'\{N/x\}))$
  $\leq \max(\omega_m(M), \omega_m(M'), \omega_m(N), size(A))$
  $\leq \max(\dots)$

## Exercise 8.

By induction on $\Gamma \vdash M : A$,

- case $\Gamma, x : A, \Delta \vdash x : A$, then $\omega_m(x) = 0$.
- case $\Gamma \vdash \lambda x.M : A \multimap B$, then $\Gamma, x : A \vdash M : B$,

  if $\omega_m(M) = 0$ then $\omega_m(\lambda x.M) = 0$

  if $\omega_m(M^*) < \omega_m(M)$ then

$$\omega_m((\lambda x.M)^*) = \omega_m(\lambda x.M^*)$$
$$= \omega_m(M^*)$$
$$< \omega_m(M)$$
$$= \omega_m(\lambda x.M)$$

- case $\Gamma \vdash MN : A$ then $\Gamma \vdash M : B \multimap A$ and $\Gamma \vdash N : B$.

if $M = \lambda x. L$, then $(MN)^* = L^* \{N^*/x\}$

and thus

$$\omega_m((MN)^*) = \omega_m(L^* \{N^*/x\}) \leq \max(\omega_m(L^*), \omega_m(N^*), size(A))$$

$$\omega_m(MN) = \omega_m((\lambda x\, L)\, N) \leq \max(\omega_m(L), \omega_m(N), size(A))$$

and we conclude by IH on $L$ and $N$.

otherwise $\omega(M^* N^*) \leq \max(\omega_m(M^*), \omega_m(N^*))$

$\omega(M\ N) \leq \max(\omega_m(M), \omega_m(N))$

and we conclude.

# Tutorial n° 7

## Exercise 1.

dom $\beta = \{R \mid \exists P, R \beta P\}$.

Let us show the following lemma:

<u>Lemma</u>: if $F, F'$ are CBN-contexts and $R, P \in \text{dom } \beta \cup \text{dom } \mathcal{C} \cup \text{dom}_{cc}$, such that $F[R] = F'[P]$ then $R = P$ and $F = F'$.

<u>Proof</u> by induction on $F, F'$

- if $F = F' = [\,]$ then $F[R] = R = P = F'[P]$
- if $F[R]\ N = F'[P]\ M$ then we must have $N = M$ and $F[R] = F'[P]$, and we conclude by i.h.
- if $F'[R] = R = F[P]\ N$

  then $R = (\lambda x. R')\ R''$ or $R = \mathcal{C}\ R'$ or $R = cc\ R'$
  and $P = (\lambda x. P')\ P''$ or $P = \mathcal{C}\ P'$ or $P = cc\ P'$

  if $F = [\,]$, then impossible
  if $F = F''[\,]\ M$, then impossible.

For CBV, do it next time.

$\lambda x.\lambda y.x$ and $\lambda x.\lambda y.y$ are not $\xleftrightarrow[*]{\varepsilon\, cc\, \mathcal{C}}$.

## Exercise 3.

Q1. $M \left\{ {}^{[\alpha](LN)} \big/ {}_{[\alpha]L} \right\} :=$

$$\begin{cases} [\alpha]\,(L\{\cdots\}N) & \text{if } M = [\alpha]\,L \\ [\beta]\,L\{\cdots\} & \text{if } M = [\beta]\,L \text{ with } \beta \neq \alpha \\ \mu\alpha.L & \text{if } M = \mu\alpha.L \\ \mu\beta.L\{\cdots\} & \text{if } M = \mu\beta.L \text{ with } \beta \neq \alpha \\ \lambda x.L\{\cdots\} & \text{if } M = \lambda x.L \\ x & \text{if } M = x \\ L\{\cdots\}\ R\{\cdots\} & \text{if } M = LR \end{cases}$$

Q2. For $\lambda$-calculus, the substitution lemma is:
if $\Gamma \vdash N : A$ and $\Gamma, x : A \vdash M : B$ then $\Gamma \vdash M\{N/x\} : B$.

Now, if $\Gamma \vdash N : B \mid \Delta$ and $\Gamma \vdash M : C \mid \Delta, \alpha : B \Rightarrow A$ (*)

then $\Gamma \vdash M \left\{ {}^{[\alpha]LN} \big/ {}_{[\alpha]L} \right\} : C \mid \Delta, \alpha : A$.

We can show it by induction on (*), we only need to consider the yellow cases.

For the case $\mu\alpha.M$, then we must use rule $\mu$ for (*) which is impossible as the premise with two $\alpha$s, absurd!

For the $[\alpha]M$ case then

$$\frac{}{\Gamma \vdash \underbrace{([\alpha]M)\{\cdots\}}_{[\alpha]\,(M\{\cdots\})\,N} : C \mid \Delta, \alpha : A}$$

TD n° 10

## Exercise 68 from the notes.

By induction on $A$,

- if $A = X$, then as $M' \sim_X N\,[\eta : \tau \leftrightarrow \tau']$
  we have $(M', N) \in \eta(X)$
  and by admissibility,

$$M \cong_{\tau(X)} M' \;\eta(X)\; N \cong_{\tau'(X)} N$$

implies $M \sim N\,[\eta : \tau \leftrightarrow \tau']$

- if $A = B \Rightarrow C$, then take $L \sim_B P\,[\eta : \tau \leftrightarrow \tau']$

  and we need to prove $M'L \sim_C NP\,[\eta : \tau \leftrightarrow \tau']$.

  By hypothesis and context closure, $[\,]L$

$$ML \cong_{\tau(C)} M'L \sim_B NP\,[\eta : \tau \leftrightarrow \tau']$$

  thus by induction hypothesis, we conclude

$$ML \sim_C NP\,[\eta : \tau \leftrightarrow \tau'].$$

- if $A = \forall X, B$, take $C, C'$ and $R : C \leftrightarrow C'$,

  we want to show that

$$M'C \sim_B NC'\,[\underbrace{\eta[X \mapsto R] : \tau[X \mapsto C] \leftrightarrow \tau'[X \mapsto C']}_{\eta'}]$$

  by hypothesis, $M \cong_{\tau(\forall X, B)} N$ thus, by context closure $([\,]c)$

$$MC \cong_{\tau(B)[C/X]} M'C \sim_B NC'\,[\eta']$$

  and we conclude by induction hypothesis.

Corollary: Given a closed type $A$, then

$$M \sim_A N \quad \text{iff} \quad M \cong_A N.$$

Exercise (Proof of Corollary 2.3.10):

- $M \cong_{A \Rightarrow B} N$ iff $M \sim_{A \Rightarrow B} N$

  iff $\forall U \sim_A V, \quad MU \sim_B NV$

  iff $\forall U \cong_A V, \quad MU \cong_B NV$

- if $M \cong_{\forall X, B} N$ we have by context closure $MC \cong_{B\{C/X\}} NC$. for all $C$
- if $\forall C, \ MC \cong_{B\{C/X\}} NC$, then $MC \sim_{B\{C/X\}} NC \ \forall C$,

  Let $R : D \leftrightarrow D'$, let us show $M \sim_B N [X \mapsto R]$,

  by the first lemma,

  $$MD \cong_{B\{D/X\}} N D \sim_B N D' [X \mapsto R]$$

  which implies the result. It remains to show

  that $\quad ND \sim_B ND' \ [X \mapsto R]$

  which is true as $N \sim_{\forall X, B} N$ implies $ND \sim_B ND' [X \mapsto R]$

Exercice 70.

Suppose $\vdash M : \forall X, (X \Rightarrow X)$ for some closed term $M$.

We have to show that $M \cong_{\forall X, X \Rightarrow X} \Lambda X, \lambda x\ x$

i.e. $\forall A, \ MA \cong_{A \Rightarrow A} (\Lambda X. \lambda x. x)\, A =_\beta \lambda a'. a'$

i.e. $\forall A\, a, \ MAa \cong_A (\lambda a'. a')\, a =_\beta a$

i.e. $\forall A\, a,\ M A\, a \cong_A a$. Let $A, a$.

By parametricity, $M \sim_{\forall X,\, X \Rightarrow X} M$.

Define $\sigma(X) := A$ and $\eta(X) := \{(P,Q) \mid P \cong_A a \cong_A Q\}$.

Then, $MA \sim_{X \Rightarrow X} MA\ [\eta]$ and thus $MA\, a \sim_X MA\, a\ [\eta]$

(as $a \sim_X a\ [\eta]$)

which implies $MA\, a\ \eta(A)\ MA\, a$ which implies $MA\, a \cong_A a$.

Exercise. Consider $A := \forall X\, Y,\ X \Rightarrow (X \Rightarrow Y) \Rightarrow Y$.

Prove that if $M : A$ then $M \cong_A \Lambda X Y, \lambda x f\, f x$.

Let $B, C$, $U : B$ and $V : B \Rightarrow C$. Let us show

$$M\ B\ C\ U\ V \cong_C V\, U.$$

By parametricity, we have $M \sim_A M$, thus

$\sigma := \begin{pmatrix} X \longmapsto B \\ Y \longmapsto C \end{pmatrix}$ and $\eta(X) := \{(P,Q) \mid P \cong_B U \cong_B Q\}$.

and $\eta(Y) := \{(P,Q) \mid P \cong_C V\, U \cong_C Q\}$.

this implies

$$M B C U V \sim_Y M B C U V\ [\eta]$$

which implies $MBCUV \cong_C VU$, as

$V \sim_{X \Rightarrow Y} V\ [\eta]$ as

Exercise: Given $M$ a closed term of type Bool, show

$$M \cong_{Bool} \text{False} \quad \text{or} \quad M \cong_{Bool} \text{True}.$$

i.e. by ext, for all $A$, $tt$, $ff$, for all $A'$, $tt'$, $ff'$

$$M\ A\ tt\ ff \cong tt \quad \text{or} \quad ff$$

By parametricity, $M \sim_{Bool} M$.

Define $\sigma(X) := A$ and $\eta(X) := \left\{ (P, Q) \;\middle|\; P \cong_A Q \text{ and } \left( \begin{matrix} P \cong tt \\ \text{or} \\ P \cong ff \end{matrix} \right) \right\}$.

Then, $M\ A \sim M\ A\ [\eta]$

then $M\ A\ tt \sim M\ A\ tt\ [\eta]$ as $tt \sim_X tt\ [\eta]$

then $M\ A\ tt\ ff \sim M\ A\ tt\ ff\ [\eta]$ as $ff \sim_X ff\ [\eta]$

$$\text{pair} := \lambda ab.\, \underbrace{\Lambda X. \lambda f.\, f\,a\,b.}_{\langle a,b\rangle}$$

(1) if $M : A \bar{\wedge} B$, then there exists $P : A$, $Q : B$,

$$M\,(A \bar{\wedge} B)\ \text{pair} \cong \text{pair}\ P\,Q.$$
$$=_\beta \Lambda X. \lambda f.\, f\,P\,Q$$

By parametrecity,

$$M \sim_{A \bar{\wedge} B} M$$

then define $\sigma(X) := A \bar{\wedge} B$ $\qquad \eta(X) := \left\{ (H, K) \,\middle|\, \begin{array}{l} \exists P, Q, \\ H \cong K \cong \langle P, Q \rangle \end{array} \right\}$

Then $M\,(A \bar{\wedge} B) \sim_{(A \Rightarrow B \Rightarrow X) \Rightarrow X} M\,(A \bar{\wedge} B)$ $\qquad = \bigcup\limits_{\substack{P : A \\ Q : B}} [\langle P, Q \rangle]^2_{\cong}$

$M\,(A \bar{\wedge} B)\ \text{pair} \sim_X M\,(A \bar{\wedge} B)\ \text{pair}$

and we have

$$\text{pair} \sim_{A \Rightarrow B \Rightarrow X} \text{pair}$$

i.e. $\forall U \sim U'$, $\forall V \sim V'$,

$$\text{pair}\ U\ V \sim \text{pair}\ U'\ V'$$

# TD n° 12.

## Exercise 2.

We will show that $M \cong \langle \pi_1 M, \pi_2 M \rangle$.

By parametrecity, we have that

$$M \sim_{A_1 \bar{\wedge} A_2} M$$

and so, defining $\sigma(X) := A_1 \bar{\wedge} A_2$ and $\eta(X) := \left\{ (K, K') \;\middle|\; \begin{array}{l} K \cong K' \\ \cong \langle \pi_1 K, \pi_2 K \rangle \end{array} \right\}$,

we have that

$$M(A_1 \bar{\wedge} A_2) \sim_{(A_1 \Rightarrow A_2 \Rightarrow X) \Rightarrow X} M(A_1 \bar{\wedge} A_2)\ [\eta : \sigma \leftrightarrow \sigma]$$

as $\quad pair_{A_1, A_2} \sim_{A_1 \Rightarrow A_2 \Rightarrow X} pair_{A_1, A_2} \quad [\eta : \sigma \leftrightarrow \sigma]$

i.e. $\forall P \sim_{A_1} P', \forall Q \sim_{A_2} Q'$, $\underset{\substack{\|_\beta \\ \langle P, Q \rangle}}{pair\ P\ Q} \sim_X \underset{\substack{\|_\beta \\ \langle P', Q' \rangle}}{pair\ P'\ Q'}\ [\eta : \sigma \leftrightarrow \sigma']$,

$$\langle P, Q \rangle \sim_X \langle P', Q' \rangle [\eta]$$

i.e. $\forall P \cong_{A_1} P'$, $\forall Q \cong_{A_2} Q'$, $\langle P, Q \rangle \underset{\text{by context closure}}{\cong} \langle P', Q' \rangle \cong \left( \pi_1 \langle P, Q \rangle, \pi_2 \langle P, Q \rangle \right)$.

by $=_\beta$

Thus we have that

$$H := M(A_1 \bar{\wedge} A_2)\ pair \sim_X M(A_1 \bar{\wedge} A_2)\ pair\ [\eta : \sigma \leftrightarrow \sigma]$$

i.e.

$$H \cong H \cong \langle \pi_1 H, \pi_2 H \rangle.$$

It then suffices to show that $H \cong M$, i.e. by extensionality,

$$\forall B, \forall N : A_1 \Rightarrow A_2 \Rightarrow B, \quad H\ B\ N \cong_B M\ B\ N.$$

By parametricity, $M \sim_{A_1 \bar{\wedge} A_2} M$, and defining $\sigma(X) := B$, $\sigma'(X) := A_1 \bar{\wedge} A_2$

and $\eta(X) := \{(P,Q) \mid P \simeq Q\, B N\}$.

we have $M\, B\, N \sim_X M\,(A_1 \bar{\wedge} A_2)\, \text{pair} = M\,[\eta : \sigma \leftrightarrow \sigma']$

which immediatly concludes.

We have to show that $N \sim_{A_1 \Rightarrow A_2 \Rightarrow X} \text{pair}$, i.e. $\forall P \sim_{A_1} P', \forall Q \sim_{A_2} Q',$

$$N\; P\; Q \sim_X \underset{\substack{\|_\beta \\ \langle P', Q' \rangle}}{\text{pair}\; P' Q'}\; [\eta : \sigma \leftrightarrow \sigma']$$

i.e. $N\, P\, Q \simeq \langle P', Q' \rangle\, B\, N =_\beta N\, P'\, Q'$

and we have that by context closure.

Exercise 1.

$$\boxed{\begin{array}{l} cc : ((A \Rightarrow B) \Rightarrow A) \Rightarrow A \\ \mathcal{C} : ((A \Rightarrow B) \Rightarrow B) \Rightarrow A \end{array}}$$

(with $B = \bot$)

$\mathcal{A} : \bot \Rightarrow A$.

Define $cc' := \lambda f^{:(A \Rightarrow B) \Rightarrow A}\, \mathcal{C}(\lambda g^{:A \Rightarrow B}\, g\,(f\; g))$.

$$E[\,cc'\; M] \longrightarrow_\beta E[\,\mathcal{C}(\lambda g.\; g\,(M\; g))] \xrightarrow{\mathcal{C}^{CBN}} (\lambda g.\; g\,(M\, g))\,(\lambda z. E[z]).$$
$$\longrightarrow_\beta (\lambda z. E[\,z\,])\,(M\,(\lambda z. E[z]))$$
$$\longrightarrow_\beta E[M\,(\lambda z. E[z])].$$

Define $\mathcal{C}' := \lambda f^{:(A \Rightarrow B) \Rightarrow B}\; cc\,(\lambda g.\, \mathcal{A}\, f\; g)$.

$$E[\,\mathcal{C}' M] \longrightarrow E[\,cc\,(\lambda g. \mathcal{A}\; M g)] \longrightarrow^* E[\,\mathcal{A}(M\,(\lambda z. E[z]))]$$
$$\longrightarrow M\,(\lambda z.\, E[z])$$

Exercise 3.

Define $\mathrm{case}^{A_1 \bar{\vee} A_2} : \forall X.\ A_1 \bar{\vee} A_2 \Rightarrow (A_1 \to X) \Rightarrow (A_2 \Rightarrow X) \Rightarrow X$
$:= \Lambda X.\ \lambda p\ a\ b.\ p\ X\ a\ b.$

$\mathrm{case}\ (A_1 \bar{\vee} A_2)\ (in_i\ U)\ in_1\ in_2 \longrightarrow_\beta^* (in_i\ U)(A_1 \bar{\vee} A_2)\ in_1\ in_2$
$\longrightarrow_\beta^* in_i\ U.$

Exercise 4.

Suppose $\exists M : A_1\ \exists N : A_2,\quad in_1\ M \cong in_2\ N$

and thus by context closure

$\mathrm{case}\ \mathrm{Bool}\ (\lambda\_.\ \mathrm{True})\ (\lambda\_.\ \mathrm{False})\ (in_1\ M) \cong \mathrm{case}\ \mathrm{Bool}\ (\lambda\_.\ \mathrm{True})\ (\lambda\_.\ \mathrm{False})\ (in_2\ N)$

$\quad \|_\beta \qquad\qquad\qquad \|_\beta$

$\mathrm{True} \cong \mathrm{False}$

which is absurd.

Exercise 5.

By parametricity, $M \sim_{A_1 \bar{\vee} A_2} M$

Thus $M(A_1 \bar{\vee} A_2) \sim_{(A_1 \Rightarrow X) \Rightarrow (A_2 \Rightarrow X) \Rightarrow X} M(A_1 \bar{\vee} A_2)\ [\eta : \tau \leftrightarrow \tau]$

Define $\tau(X) := A_1 \bar{\vee} A_2 \qquad \eta(X) := \{(P,Q) \mid \exists R,\ P \cong Q \cong in_1 R\}$
$\cup \{(P,Q) \mid \exists R,\ P \cong Q \cong in_2 R\}$

$M\ (A_1 \bar{\vee} A_2)\ \ in_1\ in_2 \sim_X M\ (A_1 \bar{\vee} A_2)\ \ in_1\ in_2\ [\eta : \tau \leftrightarrow \tau].$

which then implies that either $M\ (A_1 \bar{\vee} A_2)\ \ in_1\ in_2 \cong in_1 R$ for some R.
or $M\ (A_1 \bar{\vee} A_2)\ \ in_1\ in_2 \cong in_2 R$ for some R

We only have to show that

$in_i \sim_{A_i \Rightarrow X} in_i\ [\eta : \tau \leftrightarrow \tau]$

i.e. $\forall U \sim_{A_i} V,\ \ in_i\ U \sim_X in_i\ V\ [\eta : \tau \leftrightarrow \tau].$

i.e. $\forall U \simeq_{A_i} V,\ \ in_i\ U \cong in_i\ V$ with $R = U$

↳ by context-closure.

Then, it remains to show that

$$M \cong_{A_1 \bar{\vee} A_2} M\ (A_1 \bar{\vee} A_2)\ \ im_1\ im_2$$

i.e. by ext,

$$\forall B,\ \begin{array}{l} U : A_1 \Rightarrow B \\ V : A_2 \Rightarrow B \end{array} \quad M\ B\ U\ V \cong_B M\ (A_1 \bar{\vee} A_2)\ \ im_1\ im_2\ B\ U\ V.$$

By parametricity,

$$M \sim_{A_1 \bar{\vee} A_2} M$$

and, defining $\quad \tau(X) := B \qquad \tau'(X) := A_1 \bar{\vee} A_2$

and $\eta(X) := \{ (P, Q) \mid P \cong Q\ B\ U\ V \}$,

we have

$$M\ B \sim_{(A_1 \Rightarrow X) \Rightarrow (A_2 \Rightarrow X) \Rightarrow X} M(A_1 \bar{\vee} A_2) \quad [\eta : \tau \leftrightarrow \tau']$$

and $\quad M\ B\ U\ V \sim_B M(A_1\ \bar{\vee} A_2)\ im_1\ im_2 \quad [\eta]$

as $U \sim_{A_1 \Rightarrow X} im_1\ [\eta]$ i.e $\forall R \cong_{A_1} S$,

$$UR \cong (im_1 S)\ B\ U\ V$$
$$=_\beta$$
$$U\ S$$

by context closure.

and thus $M\ B\ U\ V \cong_B M(A_1\ \bar{\vee} A_2)\ im_1\ im_2$